

SLEEP



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Sleep

Sleep is the natural periodic state of rest for mind and body with closed eyes characterized by partial or complete loss of consciousness.

Loss of consciousness leads to decreased response to external stimuli and decreased body movements.

It is to be distinguished from coma, which is unconsciousness from which a person cannot be aroused.

Depth of sleep is not constant throughout the sleeping period.

It varies in different stages of sleep

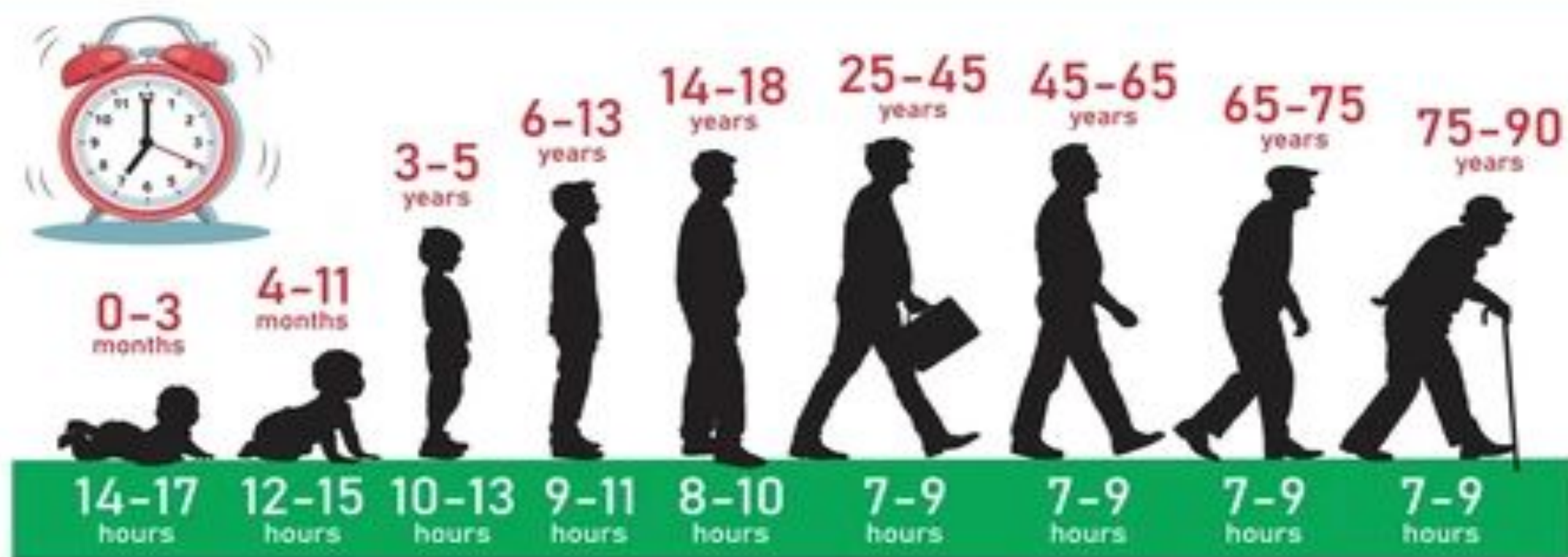


SLEEP REQUIREMENT FOR THE HUMANS

Sleep requirement is not constant. However, average sleep requirement per day at different age groups is:

1. Newborn infants : 18 to 20 hours
2. Growing children : 12 to 14 hours
3. Adults : 7 to 9 hours
4. Old persons : 5 to 7 hours

SLEEP TIME INFOGRAPHIC



HEALTH BENEFITS OF SLEEP



Supports
Heart Health



Supports Healthy
Weight Management



Improves
Physical Recovery



Helps Regulate
Blood Sugar Levels



Enhances Memory
and Brain Function



Enhances
and Brain
Function

Supports
Healthy Weight
Management



Increases Daytime
Energy and Productivity



Enhances
Balance and
Coordination

Promotes Emotional Resilience

PHYSIOLOGICAL CHANGES DURING SLEEP

During sleep, most of the body functions are reduced to basal level. Following are important changes in the body during sleep:

1- PLASMA VOLUME

Plasma volume decreases by about 10% during sleep

2- Cardiovascular System

Heart Rate: During sleep, the heart rate reduces. It varies between 45 and 60 beats per minute.

Blood Pressure: Systolic pressure falls to about 90 to 110 mm Hg. Lowest level is reached about 4th hour of sleep and remains at this level till a short time before waking up. Then, the pressure commences to rise.

If sleep is disturbed by exciting dreams, the pressure is elevated above 130 mm Hg.

3- RESPIRATORY SYSTEM: Rate and force of respiration are decreased. Respiration becomes irregular and Cheyne-Stokes type of periodic breathing may develop.

4- GASTROINTESTINAL TRACT: Salivary secretion decreases during sleep. Gastric secretion is not altered or may be increased slightly. Contraction of empty stomach is more vigorous.

5- EXCRETORY SYSTEM: Formation of urine decreases and specific gravity of urine increases.

6- SWEAT SECRETION: Sweat secretion increases during sleep.

7- LACRIMAL SECRETION: Lacrimal secretion decreases during sleep.

8-MUSCLE TONE: Tone in all the muscles of body except ocular muscles decreases very much during sleep. It is called sleep paralysis.

9- REFLEXES: Certain reflexes particularly knee jerk, are abolished. Babinski sign becomes positive during deep sleep. Threshold for most of the reflexes increases. Pupils are constricted. Light reflex is retained. Eyeballs move up and down.

10- BRAIN: Brain is not inactive during sleep. There is a characteristic cycle of brain wave activity during sleep with irregular intervals of dreams. Electrical activity in the brain varies with stages of sleep

TYPES OF SLEEP

Sleep is of two types:

- 1. Rapid eye movement sleep or REM sleep**
- 2. Non-rapid eye movement sleep, NREM sleep or non-REM sleep.**

1. RAPID EYE MOVEMENT SLEEP – REM SLEEP

Rapid eye movement sleep is the type of sleep associated with rapid conjugate movements of the eyeballs, which occurs frequently. Though the eyeballs move, the sleep is deep. So, it is also called paradoxical sleep.

It occupies about 20% to 30% of sleeping period.

This stage usually begins about 90 minutes after you fall.

Functionally, REM sleep is very important because, it plays an important role in consolidation of memory. Dreams occur during this period.

2. NON-RAPID EYE MOVEMENT SLEEP – NREM OR NON-REM SLEEP

Non-rapid eye movement (NREM) sleep is the type of sleep without the movements of eyeballs. It is also called slow-wave sleep. Dreams do not occur in this type of sleep and it occupies about 70% to 80% of total sleeping period.

Non-REM sleep is followed by REM sleep.

STAGES OF SLEEP AND EEG PATTERN

RAPID EYE MOVEMENT SLEEP

During REM sleep, electroencephalogram (EEG) shows irregular waves with high frequency and low amplitude. These waves are desynchronized waves.

NON-RAPID EYE MOVEMENT SLEEP

The NREM sleep is divided into four stages, based on the EEG pattern. During the stage of wakefulness, i.e. while lying down with closed eyes and relaxed mind, the alpha waves of EEG appear. When the person proceeds to drowsy state, the alpha waves diminish .

Stage I: Stage of Drowsiness

Alpha waves are diminished and abolished. EEG shows only low voltage fluctuations and infrequent delta waves.

Stage II: Stage of Light Sleep

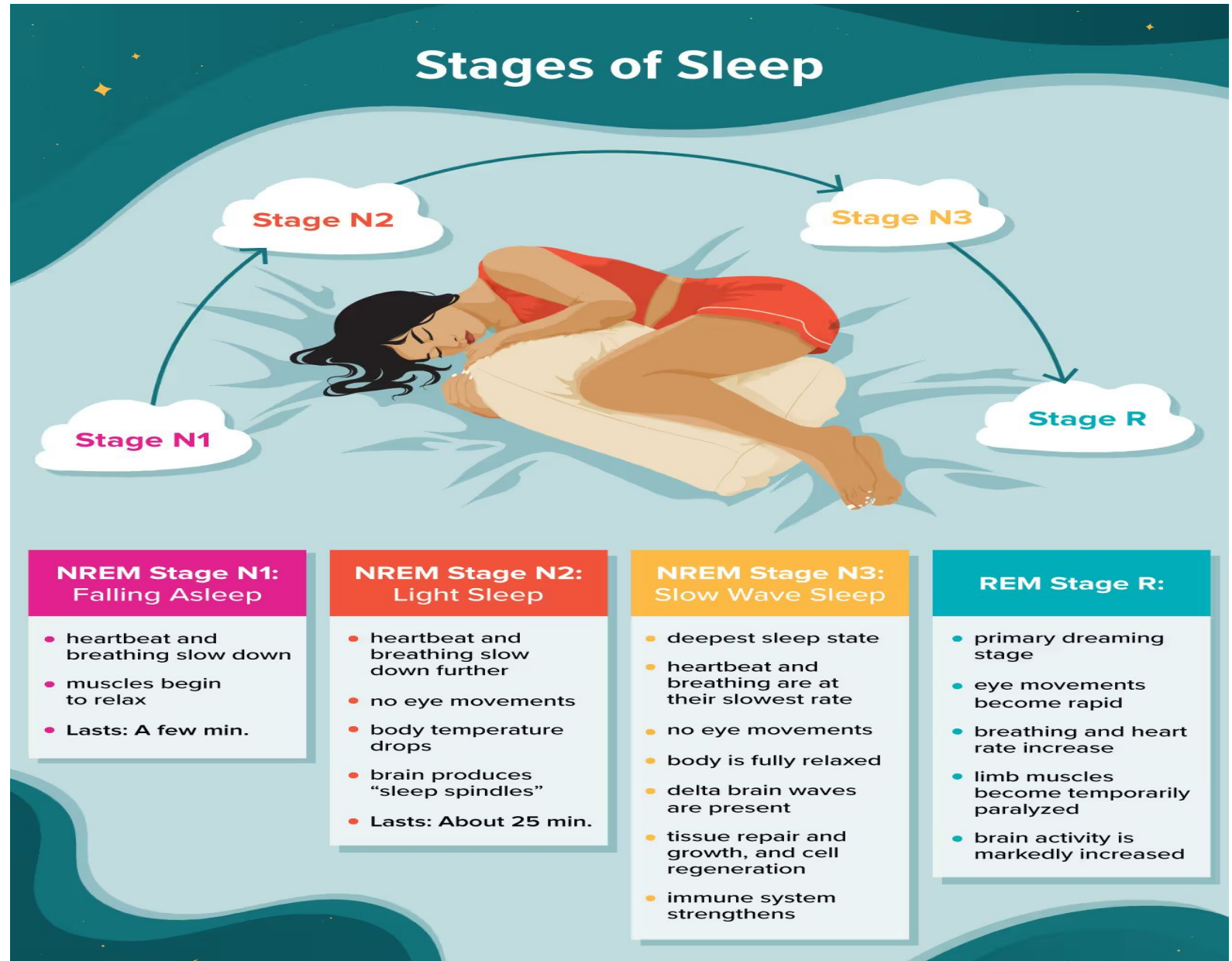
Stage II is characterized by spindle bursts at a frequency of 14 per second, superimposed by low voltage delta waves

Stage III: Stage of Medium Sleep

During this stage, the spindle bursts disappear. Frequency of delta waves decreases to 1 or 2 per second and amplitude increases to about 100 μV .

State IV: Stage of Deep Sleep

Delta waves become more prominent with low frequency and high amplitude.



MECHANISM OF SLEEP

Sleep occurs due to the activity of some sleep-inducing centers in brain. Stimulation of these centers induces sleep. Damage of sleep centers results in sleeplessness or persistent wakefulness called insomnia.

SLEEP CENTERS

Complex pathways between the reticular formation of brainstem, diencephalon and cerebral cortex are involved in the onset and maintenance of sleep.

However, two centers which induce sleep are located in brainstem:

1. Raphe nucleus
2. Locus coeruleus of pons

1. Role of Raphe Nucleus

Raphe nucleus is situated in lower pons and medulla. Activation of this nucleus results in non-REM sleep. It is due to release of serotonin by the nerve fibers arising from this nucleus. Serotonin induces non-REM sleep.

2. Role of Locus Coeruleus of Pons

Activation of this center produces REM sleep. Noradrenaline released by the nerve fibers arising from locus coeruleus induces REM sleep

Inhibition of Ascending Reticular Activating System

Ascending reticular activating system (ARAS) is responsible for wakefulness because of its afferent and efferent connections with cerebral cortex. Inhibition of ARAS induces sleep. Lesion of ARAS leads to permanent somnolence, i.e. coma.



SLEEP PHYSIOLOGY

Sleep physiology is regulated by a dual-process model involving a homeostatic sleep drive and a circadian rhythm that control transitions in the brain via a mutual inhibition "flip-flop switch"

Two-Process Regulation

Process S (Homeostatic Sleep Drive): Adenosine builds up in the brain during wakefulness, creating an increasing pressure to sleep that dissipates overnight.

Caffeine blocks adenosine receptors to promote alertness

Process C (Circadian Rhythm): Driven by the suprachiasmatic nucleus (SCN) in the hypothalamus, this roughly 24-hour internal clock aligns wakefulness and melatonin production from the pineal gland with the light-dark cycle

The Sleep-Wake Switch

Arousal Systems: The reticular activating system (RAS) in the brainstem and regions releasing histamine, norepinephrine, and serotonin maintain wakefulness.

Sleep Promoting: The ventrolateral preoptic nucleus (VLPO) in the hypothalamus uses GABA and galanin to inhibit arousal centers, allowing the brain to fall asleep

APPLIED PHYSIOLOGY

SLEEP DISORDERS

1. INSOMNIA

Insomnia is the inability to sleep or abnormal wakefulness. It is the most common sleep disorder. It occurs due to systemic illness or mental conditions such as psychiatric problems, alcoholic addiction and drug addiction.

2. HYPERSOMNIA

Hypersomnia is the excess sleep or excess need to sleep. It occurs because of lesion in the floor of the third ventricle, brain tumors, encephalitis, chronic bronchitis and disease of muscles. Hypersomnia also occurs in endocrine disorders such as myxedema and diabetes insipidus.

3-NARCOLEPSY AND CATAPLEXY

Narcolepsy is the sudden attack of uncontrollable sleep.

Cataplexy is sudden outburst of emotion. Both the diseases are due to hypothalamic disorders.

4. SLEEP APNEA SYNDROME

Sleep apnea is the temporary stoppage of breathing repeatedly during sleep. Sleep apnea syndrome is the disorder that involves fluctuations in the rate and force of respiration during REM sleep with short apneic episode. Apnea is due to decreased stimulation of respiratory centers, arrest of diaphragmatic movements, airway obstruction or the combination of all these factors. When breathing stops, the resultant hypercapnia and hypoxia stimulate respiration.

Sleep apnea syndrome occurs in obesity, myxedema, enlargement of tonsil and lesion in brainstem.

Common features of this syndrome are loud snoring.

5. NIGHTMARE

Nightmare is a condition during sleep that is characterized by a sense of extreme uneasiness or discomfort or by frightful dreams. Nightmare occurs due to improper food intake, digestive disorders or nervous disorders. It also occurs during drug withdrawal or alcohol withdrawal.

6-NIGHT TERROR

Night terror is a disorder similar to nightmare. It is common in children. It is also called pavor nocturnus or sleep terror. The child awakes screaming in a state of fright and semi consciousness. The child cannot recollect the attack in the morning. Nightmare occurs shortly after falling asleep and during non-REM sleep. There is no psychological disturbance.

7. SOMNAMBULISM

Somnambulism is getting up from bed and walking in the state of sleep. It is also called walking during sleep or sleep walking

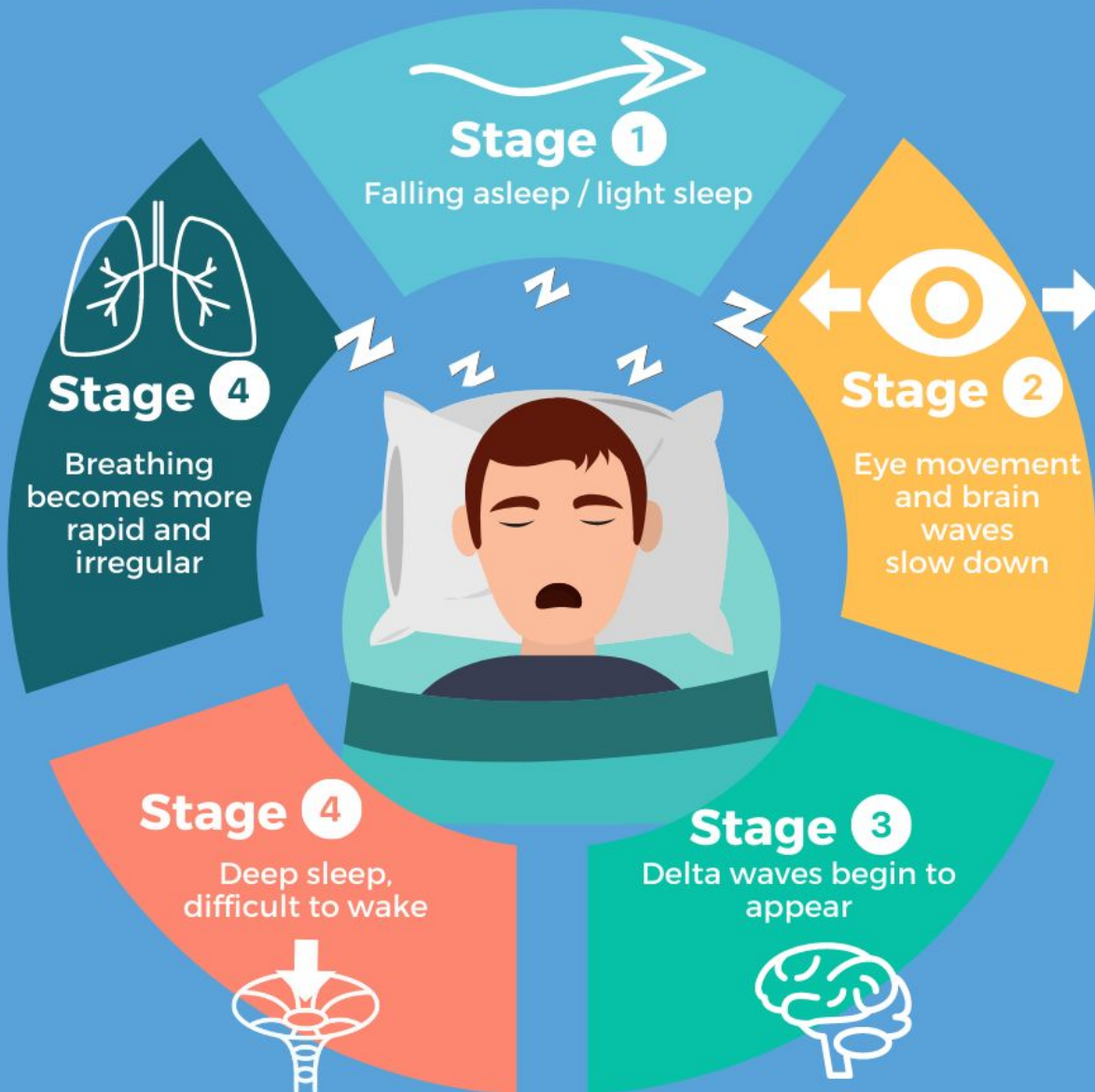
8. NOCTURNAL ENURESIS

Nocturnal enuresis is the involuntary voiding of urine at bed. It is also called or bedwetting. It is common in children.

9. MOVEMENT DISORDERS DURING SLEEP

Movement disorders occur immediately after falling asleep. Sleep start or hypnic jerk is the common movement disorder during sleep. It is characterized by sudden jerks of arms or legs. Sleep start is a physiological form of clonus. Other movement disorders are teeth grinding (bruxism), banging the head and restless movement of arms or legs

The Sleep Cycle



carenitix

5 COMMON TYPES OF SLEEP DISORDERS



1.

Insomnia

Difficulty falling asleep or staying asleep.

2.

Sleep Apnea

Breathing repeatedly starts and stops.

3.

Hypersomnia

Feeling exhausted after 8-10 hours of sleep.

4.

Narcolepsy

Irregular sleep-wake cycles.

5.

Parasomnia

abnormal or unusual behavior during sleep.

What causes sleep disorders?

A disruption to your body's cycle of sleep and daytime wakefulness causes sleep disorders. Specific things may cause this to happen, and it varies based on the type of sleep disorder you have. They may include:

- A symptom of a medical condition like heart disease, asthma, pain or a nerve condition.
- A symptom of a mental health condition like depression or anxiety disorder.
- Genetic factors (a mutation).
- A side effect of a medication.
- Working the night shift.
- Substance use before bedtime like caffeine or alcohol.
- Low levels of certain chemicals or minerals in the brain.
- An unknown cause.

Sleep disorders

Causes of sleep disorders include:



- **Neurochemical imbalances in the brain.**
- **Sleep pathways interfering with wake.**



Genetic factors.



A medication side effect.

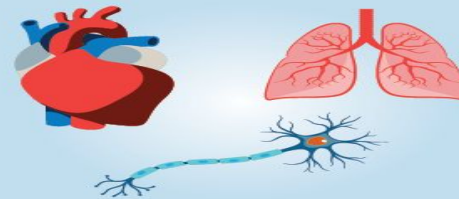


Substance use before bedtime.



Working the night shift.

A symptom of a medical condition like heart disease, asthma, pain or a nerve condition.



A symptom of a mental health condition like depression or anxiety disorder.



What happens if a person don't get enough sleep?

If a person don't get the proper amount or quality of sleep that your body needs, it can affect you beyond feeling tired during the daytime. A lack of adequate sleep can lead to:

- Difficulty learning, remembering or making decisions
- Personality changes like irritability
- Lower reaction times (making accidents more likely to happen)

Sleep loss can also contribute to the development of health conditions like:

- Depression
- Obesity
- Type 2 diabetes
- Heart disease
- Dementia

Symptoms of sleep deprivation



Excessive daytime sleepiness



Memory problems



Slower thinking and poor decision making

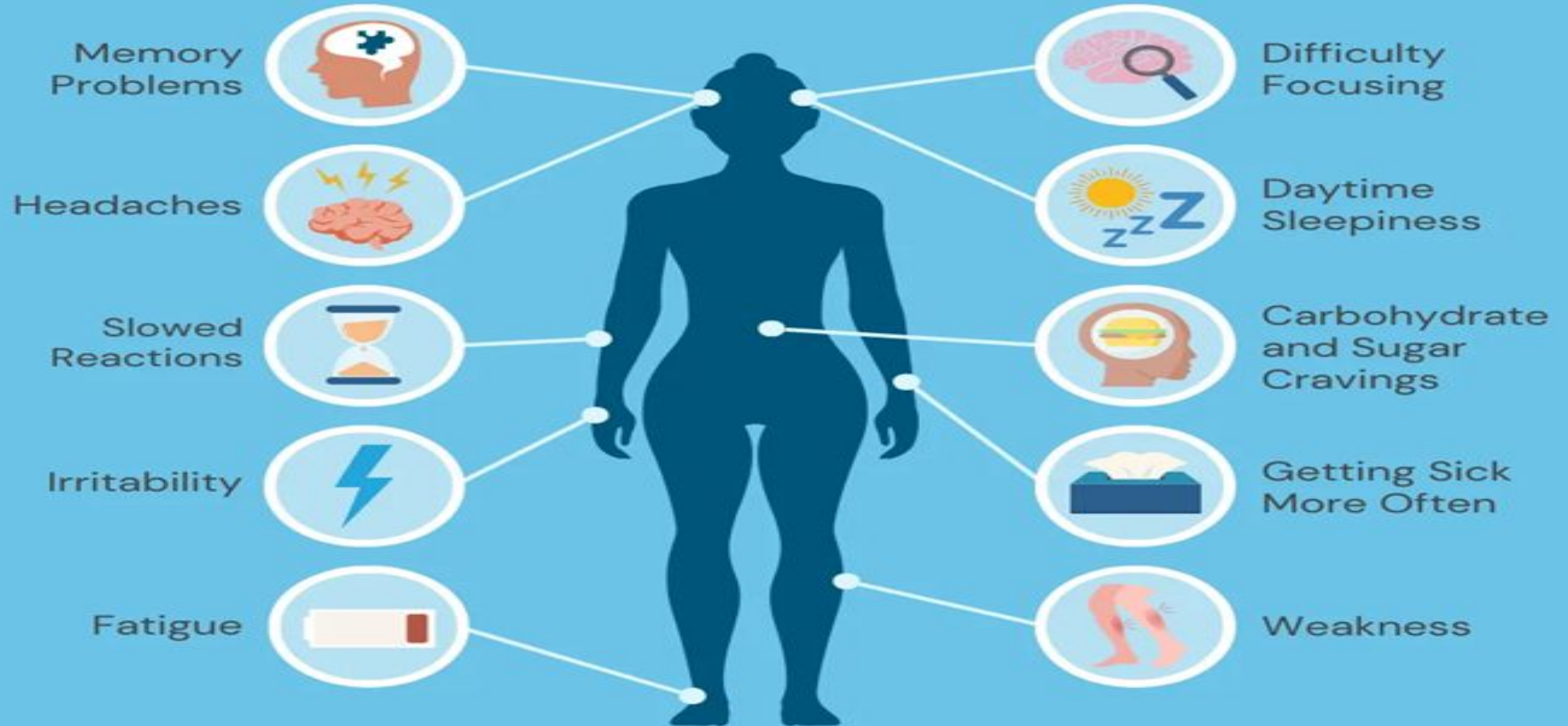


Irritability or mood swings



Difficulty concentrating

How Sleep Deprivation Affects the Body



WHY YOU SHOULD SLEEP 8HRS

8 HRS SLEEP

Faster recovery
Balanced hormones
(Testosterone,
Growth hormone)
Better focus
Improved mood
Stronger immunity
Fat loss



Faster recovery
Balanced hormones
(Growth hormone)

4 HRS SLEEP

Slower recovery
High cortisol
(Stress hormone)
Poor focus
Food cravings
Weak immunity
Fat storage



Food cravings
Weak immunity

What are the risk factors for sleep disorders?

A person may be more at risk of sleep disorders if he or she :

- Have an underlying health condition.
- Experience stress.
- Work late shifts.
- Have a history of sleep disorders in your biological family.
- Research suggests that females are more likely to experience sleep disorders than males.
- In addition, about half of all adults over the age of 65 have a type of sleep disorder.



Why Do People Need Deep Sleep?



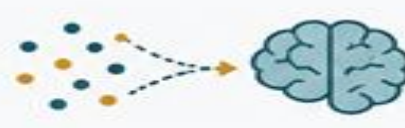
**PHYSICAL
RESTORATION**



**IMMUNE
FUNCTION**



**MEMORY
CONSOLIDATION**



**COGNITIVE
FUNCTION**



**EMOTIONAL
REGULATION**



**HORMONE
REGULATION**



**HEART
HEALTH**



What Affects Sleep Stages?

Age: Time in each stage changes dramatically over a person's life. Newborns spend far more time in REM sleep and may enter a REM stage as soon as they fall asleep. As they get older, their sleep becomes similar to that of adults. Older adults tend to spend less time in REM sleep.

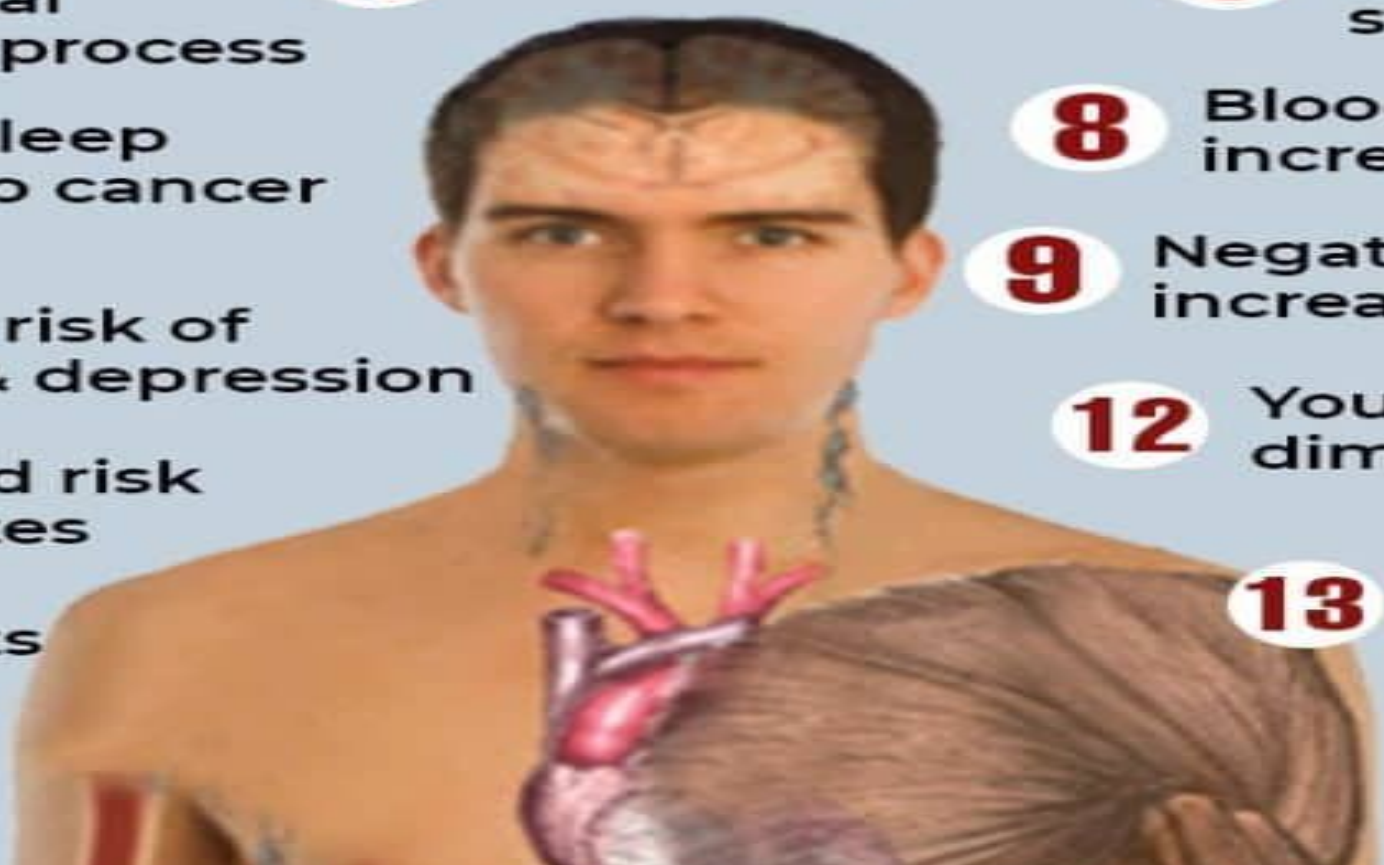
Recent sleep patterns: If a person gets irregular or insufficient sleep over a period of days or more, it can cause an abnormal sleep cycle.

Alcohol: Alcohol and some other drugs can alter sleep architecture.

For example, alcohol decreases REM sleep early in the night, but as the alcohol wears off, there's a REM sleep rebound, with prolonged REM stages.

Sleep disorders: Sleep apnea, restless legs syndrome (RLS), and other conditions that cause multiple awakenings may interrupt a healthy sleep cycle.

13 Terrifying Health Effects of Sleep Deprivation

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- 1 Breakdown of the immune system
 - 2 Breakdown of the logical thought process
 - 3 Pile up of weight & obesity risks
 - 4 Pale & dry skin
 - 5 Risk of heart disease & stroke
 - 6 Lack of sleep related to cancer risk
 - 7 Increase risk of anxiety & depression
 - 8 Blood pressure increases
 - 9 Negative thinking increases
 - 10 Increased risk of diabetes
 - 11 Body gets colder
 - 12 Your libido diminishes
 - 13 Muscle growth slows down

How do I get better sleep?

- Creating a comfortable sleep environment
- Minimizing stress
- Avoid using your bed for anything other than sleep and intimate relations:
Don't watch television or videos on your phone, eat or work in your bedroom.
- Establishing a regular bedtime routine
- Not watching the clock
- Exercising regularly

THANK YOU